MAS 812: Nonparametric Methods

Lecture 6: Tests for Several Samples

Dr. W. Arori

Learning Objectives

By the end of this lecture, you will be able to:

- Apply the Kruskal-Wallis test for comparing k independent samples
- Perform the Friedman test for k related samples (repeated measures)
- Conduct post-hoc pairwise comparisons following significant omnibus tests
- Apply the Jonckheere-Terpstra test for ordered alternatives
- Understand the median test for several samples
- Implement multiple comparison corrections (Bonferroni, Dunn, Nemenyi)
- Interpret results and choose appropriate tests for different designs

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1 Introduction to Multiple Sample Problems

When comparing three or more groups, we need to extend the two-sample methods covered in previous lectures. The design can be either independent samples (between-subjects) or related samples (within-subjects/repeated measures) [2].

Types of Designs

- **Independent samples ($k > 2$ groups):** Each subject belongs to one group only. Examples: comparing four different drugs, three teaching methods.
- **Related samples (repeated measures):** Same subjects measured under k different conditions. Examples: patients measured at 3 time points, same subjects tested under 4 different temperatures.
- **Randomized block design:** Subjects are matched into blocks, with each block receiving all treatments.

Parametric Approach: One-Way ANOVA

The classical parametric approach assumes normality and homogeneity of variances:

$$F = \frac{\text{Between-group variability}}{\text{Within-group variability}} \sim F_{k-1,N-k}$$

However, when assumptions are violated, nonparametric alternatives are needed [6].

2 Kruskal-Wallis Test

The Kruskal-Wallis test is a nonparametric alternative to one-way ANOVA, extending the Wilcoxon rank-sum test to more than two groups [2][6].

Assumptions

1. The k samples are independent random samples from their respective populations
2. The variable of interest is continuous (or at least ordinal)
3. The distributions have the same shape (for testing location differences)
4. Observations are independent within and between samples

Hypotheses

$$H_0 : \text{All } k \text{ populations have the same location (median)}$$

$$H_1 : \text{At least one population tends to yield larger values than at least one other}$$

Equivalently, H_0 states that all k distribution functions are identical [9].

Test Procedure

1. Combine all $N = n_1 + n_2 + \cdots + n_k$ observations
2. Rank all observations from smallest (rank 1) to largest (rank N), assigning average ranks for ties
3. Calculate the rank sum R_i for each group $i = 1, \ldots, k$
4. Compute the Kruskal-Wallis test statistic H [2]:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

5. Under H_0, H follows approximately a chi-square distribution with $k-1$ degrees of freedom, provided that sample sizes are not too small (typically each $n_i \geq 5$)

Adjustment for Ties

When ties are present, a correction factor is applied:

$$H_c = \frac{H}{1 - \frac{\sum_{j=1}^{g}(t_j^3 - t_j)}{N^3 - N}}$$

where g is the number of tied groups and t_j is the size of the j-th tied group.

Example 2.1 (Kruskal-Wallis Test)**.** A researcher wants to compare the effectiveness of three different pain relievers. Patients are randomly assigned to one of three drugs, and their pain relief times (in minutes) are recorded:

Drug A: 20, 25, 22, 28, 24 Drug B: 19, 21, 24, 18, 22 Drug C: 26, 23, 27, 29, 30

Test at $\alpha = 0.05$ whether the drugs differ in pain relief time.

Solution:

Step 1: State hypotheses

$$H_0 : \text{All three drugs have the same median relief time}$$

$$H_1 : \text{At least one drug differs from the others}$$

Step 2: Combine and rank all observations

Value	Group	Rank
18	B	1
19	B	2
20	A	3
21	B	4
22	A	5.5
22	B	5.5
23	C	7
24	A	8.5
24	B	8.5
25	A	10
26	C	11
27	C	12
28	A	13
29	C	14
30	C	15

Step 3: Calculate rank sums

$$\begin{aligned} R_A &= 3 + 5.5 + 8.5 + 10 + 13 = 40 \\ R_B &= 1 + 2 + 4 + 5.5 + 8.5 = 21 \\ R_C &= 7 + 11 + 12 + 14 + 15 = 59 \end{aligned}$$

$n_A = n_B = n_C = 5$, $N = 15$

Step 4: Compute H

$$\begin{aligned} H &= \frac{12}{15 \times 16}\left(\frac{40^2}{5} + \frac{21^2}{5} + \frac{59^2}{5}\right) - 3(16) \\ &= \frac{12}{240}\left(\frac{1600}{5} + \frac{441}{5} + \frac{3481}{5}\right) - 48 \\ &= 0.05\,(320 + 88.2 + 696.2) - 48 \\ &= 0.05 \times 1104.4 - 48 = 55.22 - 48 = 7.22 \end{aligned}$$

Step 5: Check for ties and adjust if necessary There are ties at 22 (A and B) and 24 (A and B). Compute correction factor:

$$\sum (t_j^3 - t_j) = (2^3 - 2) + (2^3 - 2) = (8 - 2) + (8 - 2) = 6 + 6 = 12$$

$$1 - \frac{12}{15^3 - 15} = 1 - \frac{12}{3375 - 15} = 1 - \frac{12}{3360} = 1 - 0.00357 = 0.99643$$

$$H_c = \frac{7.22}{0.99643} = 7.25$$

Step 6: Compare to critical value Degrees of freedom $= k - 1 = 2$ Critical value $\chi^2_{2,0.05} = 5.99$

Since $H_c = 7.25 > 5.99$, we reject H_0.

Step 7: Conclusion There is significant evidence that the pain relief times differ among the three drugs. The rank sums suggest Drug C tends to have the longest relief, Drug B the shortest.

3 Post-Hoc Analysis for Kruskal-Wallis

When the Kruskal-Wallis test rejects H_0, we need to determine which groups differ significantly. Several approaches are available [4][8].

Dunn's Test

Dunn's test is appropriate for pairwise comparisons following a significant Kruskal-Wallis result, especially with unequal sample sizes or ties [8].

The test statistic for comparing groups i and j is:

$$z = \frac{\bar{R}_i - \bar{R}_j}{\sqrt{\frac{N(N+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}}$$

where $\bar{R}_i = R_i/n_i$ is the mean rank for group i.

Nemenyi Test

For equal sample sizes and no ties, the Nemenyi test (Tukey-type) can be used [8]:

$$q = \frac{|\bar{R}_i - \bar{R}_j|}{\sqrt{\frac{k(k+1)}{6N}}}$$

Critical values are obtained from the Studentized range distribution.

Bonferroni Correction

A simple approach is to perform Mann-Whitney tests for all pairwise comparisons, but adjust the significance level [2][4]:

$$\alpha_{\text{adjusted}} = \frac{\alpha}{\text{number of comparisons}} = \frac{\alpha}{\binom{k}{2}}$$

For $k = 3$, there are 3 comparisons, so $\alpha_{\text{adj}} = 0.05/3 = 0.0167$.

Example 3.1 (Post-Hoc Analysis for Pain Relief Data)**.** Following the significant Kruskal-Wallis test, determine which drugs differ.

Solution using Dunn's test:

Mean ranks:

$$\bar{R}_A = 40/5 = 8.0, \quad \bar{R}_B = 21/5 = 4.2, \quad \bar{R}_C = 59/5 = 11.8$$

Standard error for pairwise comparisons:

$$SE = \sqrt{\frac{N(N+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)} = \sqrt{\frac{15 \times 16}{12}\left(\frac{1}{5} + \frac{1}{5}\right)} = \sqrt{20 \times 0.4} = \sqrt{8} = 2.828$$

Compute z-statistics:

$$\begin{aligned} z_{AB} &= \frac{|8.0 - 4.2|}{2.828} = \frac{3.8}{2.828} = 1.34 \\ z_{AC} &= \frac{|8.0 - 11.8|}{2.828} = \frac{3.8}{2.828} = 1.34 \\ z_{BC} &= \frac{|4.2 - 11.8|}{2.828} = \frac{7.6}{2.828} = 2.69 \end{aligned}$$

Using Bonferroni correction with $\alpha_{\text{adj}} = 0.0167$, critical $z = 2.39$ (from normal table).

Since $z_{BC} = 2.69 > 2.39$, only Drugs B and C differ significantly. Drug C has significantly longer pain relief than Drug B.

4 Median Test for Several Samples

The median test can be extended to multiple independent samples as a generalization of the two-sample median test [1].

Procedure

1. Find the overall median of all N observations
2. Create a $2 \times k$ contingency table:

	Group 1	Group 2	$\cdots$	Group k	Total
Above median	O_{11}	O_{12}	$\cdots$	O_{1k}	a
Below median	O_{21}	O_{22}	$\cdots$	O_{2k}	b
Total	n_1	n_2	$\cdots$	n_k	N

3. Compute the chi-square test statistic:

$$\chi^2 = \sum_{i=1}^{2} \sum_{j=1}^{k} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

   where expected frequencies $E_{ij} = \frac{\text{(row total)(column total)}}{N}$

4. Under H_0, χ^2 follows approximately a chi-square distribution with $k-1$ degrees of freedom

Remark 4.1. The median test has lower power than the Kruskal-Wallis test for location alternatives, but is more robust to outliers [1].

5 Friedman Test

The Friedman test is a nonparametric alternative to repeated measures ANOVA, extending the sign test to multiple related samples [2][3][7].

Assumptions

1. The data consist of n blocks (subjects) and k treatments (conditions)
2. Observations are measured on at least an ordinal scale
3. The blocks are independent of each other
4. Within each block, observations can be ranked

Hypotheses

$$H_0 : \text{All } k \text{ treatments have the same effect (distributions identical)}$$

$$H_1 : \text{At least one treatment tends to yield larger values than at least one other}$$

Test Procedure

1. Arrange data in a table with n rows (blocks) and k columns (treatments)
2. Within each block, rank the observations from 1 to k (smallest gets rank 1)
3. Sum the ranks for each column (treatment): $R_1, R_2, \ldots, R_k$
4. Compute the Friedman test statistic [3][10]:

$$F_r = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1)$$

5. Under H_0, F_r follows approximately a chi-square distribution with $k-1$ degrees of freedom for moderate to large n [2]

Adjustment for Ties

When ties occur within blocks, assign average ranks and apply a correction:

$$F_r = \frac{12 \sum_{j=1}^{k} R_j^2 - 3n^2 k(k+1)^2}{nk(k+1) + \frac{1}{k-1} \sum_{i=1}^{n} \sum_{j=1}^{k} (r_{ij}^3 - r_{ij})}$$

where r_{ij} are the ranks within block i.

Example 5.1 (Friedman Test)**.** A study measures reaction times (in milliseconds) for 5 subjects under three different conditions: Control, Low Stimulus, High Stimulus. Test whether the conditions affect reaction time.

Subject	Control	Low	High
1	245	256	265
2	238	251	258
3	240	249	260
4	242	253	262
5	239	250	257

Solution:

Step 1: Rank within each subject

Subject	Control	Low	High
1	1 (245)	2 (256)	3 (265)
2	1 (238)	2 (251)	3 (258)
3	1 (240)	2 (249)	3 (260)
4	1 (242)	2 (253)	3 (262)
5	1 (239)	2 (250)	3 (257)
R_j	5	10	15

Step 2: Compute F_r

$$F_r = \frac{12}{5 \times 3 \times 4}(5^2 + 10^2 + 15^2) - 3 \times 5 \times 4$$

$$= \frac{12}{60}(25 + 100 + 225) - 60 = 0.2 \times 350 - 60 = 70 - 60 = 10$$

Step 3: Compare to critical value Degrees of freedom $= k - 1 = 2$ Critical value $\chi^2_{2,0.05} = 5.99$

Since $F_r = 10 > 5.99$, we reject H_0.

Step 4: Conclusion There is significant evidence that reaction times differ across conditions. The rank sums suggest reaction times increase with stimulus level.

6 Post-Hoc Analysis for Friedman Test

Following a significant Friedman test, pairwise comparisons can be performed using the Wilcoxon signed-rank test with Bonferroni correction [2].

Procedure

1. Perform Wilcoxon signed-rank tests for each pair of conditions
2. Apply Bonferroni correction: $\alpha_{\text{adj}} = \alpha / \binom{k}{2}$
3. For $k = 3$, there are 3 comparisons, so $\alpha_{\text{adj}} = 0.05/3 = 0.0167$

Dunn's Test for Related Samples

An alternative is Dunn's test adapted for related samples:

$$z = \frac{|R_i - R_j|}{\sqrt{\frac{nk(k+1)}{6}}}$$

where R_i and R_j are the rank sums for the two conditions being compared.

Example 6.1 (Post-Hoc Analysis for Reaction Time Data)**.** Following the significant Friedman test, determine which conditions differ.

Solution using Wilcoxon signed-rank tests:

Comparing Control vs Low: Differences: 11, 13, 9, 11, 11 Wilcoxon test p-value ≈ 0.043

Comparing Control vs High: Differences: 20, 20, 20, 20, 18 Wilcoxon test p-value ≈ 0.043

Comparing Low vs High: Differences: 9, 7, 11, 9, 7 Wilcoxon test p-value ≈ 0.043

With Bonferroni correction, $\alpha_{\text{adj}} = 0.0167$. All p-values are $0.043 > 0.0167$, so no pairwise comparison reaches significance despite the overall test being significant. This illustrates the conservativeness of Bonferroni correction with small samples.

7 Jonckheere-Terpstra Test

The Jonckheere-Terpstra test is designed for situations where the k groups have a natural ordering (e.g., dose levels: low, medium, high) and we want to test for a trend [1].

Hypotheses

$$H_0 : \text{The } k \text{ distributions are identical}$$

$$H_1 : \text{The distributions are ordered in a specified direction}$$

$$(e.g., \theta_1 \leq \theta_2 \leq \cdots \leq \theta_k \text{ with at least one strict inequality})$$

Test Procedure

1. For each pair of groups (i, j) with $i < j$, compute the Mann-Whitney statistic U_{ij} counting the number of times an observation in group j exceeds an observation in group i
2. The Jonckheere-Terpstra statistic is:

$$JT = \sum_{i=1}^{k-1} \sum_{j=i+1}^{k} U_{ij}$$

3. Under H_0, JT has mean and variance:

$$\mathbb{E}[JT] = \frac{N^2 - \sum_{i=1}^{k} n_i^2}{4}$$

$$\text{Var}(JT) = \frac{N^2(2N+3) - \sum_{i=1}^{k} n_i^2(2n_i+3)}{72}$$

4. For large samples, $Z = (JT - \mathbb{E}[JT])/\sqrt{\text{Var}(JT)} \sim N(0,1)$

Example 7.1 (Jonckheere-Terpstra Test)**.** Using the pain relief data from earlier, suppose the drugs are ordered by expected potency: Drug B (weakest), Drug A (medium), Drug C (strongest). Test for an increasing trend.

Solution:

Compute pairwise Mann-Whitney statistics:

$$\begin{aligned}
&U_{BA} : \text{Count pairs where } A > B \\
&\text{Data: } B : 19, 21, 24, 18, 22 \quad A : 20, 25, 22, 28, 24 \\
&\qquad \text{Number of } A > B \text{ pairs} = 4 + 5 + 4 + 5 + 5 = 23 \quad \text{(out of 25)} \\
&U_{BA} = 23
\end{aligned}$$

$$\begin{aligned}
&U_{BC} : \text{Count pairs where } C > B \\
&\text{C: } 26, 23, 27, 29, 30 \\
&\qquad \text{Number of } C > B \text{ pairs} = 5 + 5 + 5 + 5 + 5 = 25 \quad \text{(all pairs)} \\
&U_{BC} = 25
\end{aligned}$$

$$\begin{aligned}
&U_{AC} : \text{Count pairs where } C > A \\
&\qquad \text{Number of } C > A \text{ pairs} = 5 + 5 + 5 + 5 + 5 = 25 \quad \text{(all pairs)} \\
&U_{AC} = 25
\end{aligned}$$

$$JT = U_{BA} + U_{BC} + U_{AC} = 23 + 25 + 25 = 73$$

Expected value under H_0:

$$\mathbb{E}[JT] = \frac{15^2 - (5^2 + 5^2 + 5^2)}{4} = \frac{225 - 75}{4} = \frac{150}{4} = 37.5$$

Variance:

$$\text{Var}(JT) = \frac{15^2(2 \times 15 + 3) - 3 \times 5^2(2 \times 5 + 3)}{72} = \frac{225 \times 33 - 75 \times 13}{72} = 89.58$$

$$SD = \sqrt{89.58} = 9.46$$

$$Z = \frac{73 - 37.5}{9.46} = \frac{35.5}{9.46} = 3.75$$

One-sided p-value ≈ 0.0001, providing strong evidence of an increasing trend.

8 Cochran's Q Test

Cochran's Q test is a nonparametric test for comparing three or more matched groups when the outcome is binary (e.g., success/failure, yes/no) [3].

Assumptions

1. Data are from n matched sets (blocks) of size k
2. Response is binary (0/1)
3. Blocks are independent

Test Procedure

Let X_{ij} be the response (0 or 1) for block i, treatment j. Let:

- $G_j = \sum_{i=1}^{n} X_{ij}$ be the column total for treatment j
- $L_i = \sum_{j=1}^{k} X_{ij}$ be the row total for block i
- $N = \sum_{j=1}^{k} G_j = \sum_{i=1}^{n} L_i$ be the grand total

The test statistic is [3]:

$$Q = \frac{k(k-1)\sum_{j=1}^{k}(G_j - N/k)^2}{\sum_{i=1}^{n} L_i(k - L_i)}$$

Under H_0 (treatments equally effective), Q follows approximately a chi-square distribution with $k - 1$ degrees of freedom.

Example 8.1 (Cochran's Q Test)**.** Six patients are asked whether they experienced relief (1 = relief, 0 = no relief) from three different medications. Test whether the medications differ.

Patient	Med A	Med B	Med C	L_i
1	1	0	0	1
2	0	0	1	1
3	0	0	0	0
4	1	0	1	2
5	0	1	1	2
6	0	0	1	1
G_j	2	1	4	$N = 7$

$$Q = \frac{3 \times 2 \times [(2 - 7/3)^2 + (1 - 7/3)^2 + (4 - 7/3)^2]}{1(3-1) + 1(3-1) + 0(3-0) + 2(3-2) + 2(3-2) + 1(3-1)}$$

Compute:

$$(2 - 2.333)^2 = 0.111, \quad (1 - 2.333)^2 = 1.778, \quad (4 - 2.333)^2 = 2.778$$

Sum of squares $= 0.111 + 1.778 + 2.778 = 4.667$

Numerator $= 6 \times 4.667 = 28$

Denominator $= 1 \times 2 + 1 \times 2 + 0 \times 3 + 2 \times 1 + 2 \times 1 + 1 \times 2 = 2 + 2 + 0 + 2 + 2 + 2 = 10$

$$Q = 28/10 = 2.8$$

Degrees of freedom $= 2$, critical value $\chi^2_{2,0.05} = 5.99$

Since $2.8 < 5.99$, fail to reject H_0. No evidence that medications differ.

9 Summary of Methods

Method	**Design**	**Data Type**	**Parametric Analog**
Kruskal-Wallis	Independent samples	Continuous/Ordinal	One-way ANOVA
Friedman	Related samples (repeated measures)	Continuous/Ordinal	Repeated measures ANOVA
Jonckheere-Terpstra	Independent samples with ordered groups	Continuous/Ordinal	Test for trend
Median test	Independent samples	Any scale	Test on medians
Cochran's Q	Related samples	Binary	McNemar's test (extended)

10 Choosing the Right Test

- **Kruskal-Wallis:** Default choice for comparing $k \geq 3$ independent groups when normality fails
- **Friedman:** Default for repeated measures or randomized block designs with $k \geq 3$ conditions
- **Jonckheere-Terpstra:** When groups have a natural ordering and you expect a trend
- **Median test:** When data are highly skewed or you specifically want to compare medians
- **Cochran's Q:** When outcomes are binary (success/failure) in a repeated measures design

11 Exercises

Exercise 11.1 (Kruskal-Wallis Test)**.** Four different teaching methods are compared. Test scores for students in each method:

Method 1: 82, 85, 79, 88, 84 Method 2: 75, 78, 72, 80, 74 Method 3: 88, 92, 86, 90, 89 Method 4: 70, 73, 68, 72, 71

(a) Perform a Kruskal-Wallis test at $\alpha = 0.05$

(b) If significant, conduct post-hoc pairwise comparisons using Dunn's test with Bonferroni correction

(c) Interpret your results

Exercise 11.2 (Friedman Test)**.** Six participants rate their satisfaction (1-10 scale) with three different smartphone interfaces:

Participant	Interface A	Interface B	Interface C
1	8	6	7
2	7	5	8
3	9	7	6
4	6	8	9
5	8	5	7
6	7	6	8

(a) Perform a Friedman test at $\alpha = 0.05$

(b) If significant, conduct post-hoc analysis using Wilcoxon signed-rank tests with Bonferroni correction

(c) Which interface appears best?

Exercise 11.3 (Jonckheere-Terpstra Test)**.** A study examines the effect of drug dose on pain relief. Four dose levels (0mg, 5mg, 10mg, 20mg) are tested, with 6 subjects per group. Pain relief scores (higher = better):

0mg: 2, 3, 1, 2, 3, 2 5mg: 4, 3, 5, 4, 3, 4 10mg: 5, 6, 4, 5, 6, 5 20mg: 7, 6, 8, 7, 6, 7

(a) Test for an increasing trend using the Jonckheere-Terpstra test

(b) Compare your results with a Kruskal-Wallis test

(c) Which test is more appropriate here and why?

Exercise 11.4 (Median Test for Several Samples)**.** Using the data from Exercise 1, perform a median test to compare the four teaching methods. Compare the results with the Kruskal-Wallis test. Which test do you think is more reliable?

Exercise 11.5 (Cochran's Q Test)**.** Eight patients try three different treatments for insomnia and report whether they slept better (1) or not (0):

Patient	Treatment A	Treatment B	Treatment C
1	1	0	1
2	0	1	1
3	1	0	0
4	0	0	1
5	1	1	1
6	0	0	0
7	1	0	1
8	0	1	1

(a) Perform Cochran's Q test at $\alpha = 0.05$

(b) If significant, conduct pairwise comparisons using McNemar's test with Bonferroni correction

(c) Interpret your findings

Exercise 11.6 (Power Comparison)**.** Generate data in R from three groups with distributions:

- Group 1: Normal(0,1)
- Group 2: Normal(0.5,1)
- Group 3: Normal(1,1)

with $n = 10$ per group.

(a) Perform one-way ANOVA, Kruskal-Wallis test, and median test

(b) Repeat 1000 times and compute empirical power for each test

(c) Repeat with Cauchy distributions (heavy-tailed) and compare

(d) What do you conclude about the relative power of these tests?

Exercise 11.7 (Conceptual Questions)**.** Answer briefly:

(a) Why is it important to follow a significant Kruskal-Wallis test with post-hoc comparisons rather than simply reporting the omnibus result?

(b) Explain the difference between the Kruskal-Wallis test and the Friedman test in terms of design and assumptions.

(c) Under what circumstances would you choose the Jonckheere-Terpstra test over the Kruskal-Wallis test?

(d) What is the purpose of Bonferroni correction in post-hoc analysis, and what is its limitation?

(e) Why is the median test generally less powerful than the Kruskal-Wallis test for detecting location differences?

A Statistical Tables

Critical Values for Kruskal-Wallis Test

For small samples, exact critical values should be used. For larger samples ($n_i \geq 5$), use chi-square approximation:

df	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$
2	4.605	5.991	7.378	9.210
3	6.251	7.815	9.348	11.345
4	7.779	9.488	11.143	13.277
5	9.236	11.070	12.833	15.086
6	10.645	12.592	14.449	16.812

Critical Values for Friedman Test

For small samples, exact tables are available. For $n > 5$ and $k > 4$, chi-square approximation with $k - 1$ degrees of freedom is adequate.

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